3.3 Laws of Exponents

Let us learn some rules to multiply and divide exponential numbers with the same base.

3.3.1. Multiplication of Numbers in Exponential form

Let us calculate the value of $2^3 \times 2^2$

$$\begin{aligned} 2^3 \times 2^2 &= (2\times2\times2)\times(2\times2) \\ &= 2\times2\times2\times2\times2 \\ &= 2^5 \\ &= 2^{3+2} \end{aligned}$$

We observe that the base of 2^3 and 2^2 is the same 2 and the sum of the powers is 5. Now, let us consider negative integers as the base.

$$\begin{aligned} (-3)^3 \times (-3)^2 &= \big[(-3)\times(-3)\times(-3)\big]\times\big[(-3)\times(-3)\big] \\ &= (-3)\times(-3)\times(-3)\times(-3)\times(-3) \\ &= (-3)^5 \\ &= (-3)^{3+2} \end{aligned}$$

We observe that the base of $(-3)^3$ and $(-3)^2$ is the same as (-3) and the sum of the power is 5. Similarly, $p^4 \times p^2 = (p\times p\times p\times p)\times(p\times p) = p^6 = p^{4+2}$

Now, for any non-zero integer 'a' and whole number 'm' and 'n', consider a^m and a^n. That is, $a^m = a\times a\times a\times\ldots\times a$ (m times) and $a^n = a\times a\times a\times\ldots\times a$ (n times)

So, $a^m \times a^n = a\times a\times a\times\ldots\times a$ (m times) $\times a\times a\times a\times\ldots\times a$ (n times)

$= a\times a\times a\times\ldots\times a$ ($m+n$ times) $= a^{m+n}$

Therefore, $a^m \times a^n = a^{m+n}$

This is called Product Rule of exponents.



Fig. 3.3

Try these

Simplify and write the following in exponential form.

1. $2^3 \times 2^5$
2. $p^2 \times p^4$
3. $x^6 \times x^4$
4. $3^1 \times 3^5 \times 3^4$
5. $(-1)^2 \times (-1)^3 \times (-1)^5$

Example 3.7 Simplify using Product Rule of exponents.

(i) $5^7 \times 5^3$ (ii) $3^3 \times 3^2 \times 3^4$ (iii) $25\times32\times625\times64$

Solution

(i) $5^7 \times 5^3 = 5^{7+3}$ [since, $a^m \times a^n = a^{m+n}$]

$= 5^{10}$

(ii) $3^3 \times 3^2 \times 3^4 = 3^{3+2} \times 3^4 = 3^5 \times 3^4$

$= 3^{5+4} = 3^9$

5	625
5	125
5	25
	5

2	64
2	32
2	16
2	8
2	4
	2

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